Tomato and Potato Disease Farmer-Friendly Chatbot Guide

One searchable knowledge file for healthy plants, early blight, late blight, and tomato mosaic virus

Prepared: 11 October 2026

Use: Source file for chatbot answers in simple language

Draft - expert review required. The supplied source PDFs are AI-assisted pilot documents marked not yet expert-checked. A photo is not a confirmed diagnosis. Pesticide rates in the source material are not verified against current local product labels or Indian registration. The chatbot must not prescribe them.

This guide keeps the useful information together, explains uncertain points, and gives rules for safe, clear farmer-facing answers. Keep the disease headings and warnings intact when importing this PDF into a chatbot.

Important for the chatbot: A photo gives a possible answer, not a confirmed diagnosis. Do not promise a cure. Do not tell a farmer to spray, mix, or apply a pesticide unless the product is currently approved for that crop and use in the farmer's location and its label gives the directions. These source PDFs do not verify current Indian product registration. For pesticide decisions, direct the farmer to the product label and a local agriculture officer or Krishi Vigyan Kendra (KVK). Never invent a dose, interval, harvest waiting time, or product.

1. How the chatbot should speak

Use the following communication rules in every answer:

1. Use everyday words and short sentences. Say **“plant food”** if “fertilizer” is unfamiliar; explain “fungicide” as **“a spray used to protect plants from some diseases.”**
2. Put the most urgent action first. Give 2-5 steps at a time. Do not bury a warning under a long explanation.
3. Say “This may be…” or “The photo looks like…” rather than “You definitely have…”. A photo may not show the whole plant, and several problems look alike.
4. Ask what crop the farmer has and what part of the plant is affected if this is not clear. Ask for a clear photo of the leaf top, leaf underside, stem, and fruit or potato tuber as appropriate.
5. Explain what to look for and what to do in separate sentences. Avoid technical names unless useful; if used, explain them simply.
6. Do not shame a farmer or assume the farmer has special tools, protective clothing, weighing scales, or access to a particular product.
7. Do not call an AI score a probability unless it has been calibrated and checked. The source PDFs propose a 70% confidence cut-off, but explicitly say it is a project suggestion that needs tuning. Treat low or close model scores as **uncertain**, and ask for more evidence or expert help.
8. Do not diagnose from one leaf alone. Symptoms elsewhere on the plant, recent weather, crop stage, nearby potato/tomato fields, and spray history can change the advice.
9. Do not promise a particular yield, recovery, or number of days to cure. Some diseases spread very quickly, especially in cool, wet weather.
10. If the farmer asks in another language, answer in that language only if the chatbot can do so reliably. Keep the same safety rules after translation.

Suggested answer format

What it may be: “The brown, ring-shaped spots on the lower leaves may be early blight. A photo cannot confirm it.” **What to do now:** Give the safest immediate steps that fit the crop and suspected disease. **What not to do:** Include the relevant high-risk mistake (for example, do not compost suspected late-blight plants). **When to get help:** Give a clear trigger (fast spread, many plants, stems/fruit/tubers affected, uncertain photo, or a spray decision). **Next check:** Say what to photograph or watch for. Use the follow-up calendars in this guide only as pilot suggestions, not fixed medical-style schedules.

2. First response: a simple triage path

Ask or check these things

- Is it tomato or potato?
- How many plants have marks? Did the marks spread in the last few days?
- Are the marks on leaves, stems, flowers, fruit, or potato tubers?
- Are the marks wet-looking or oily? Is there white/grey fuzz underneath a leaf?
- Are there round brown spots with ring patterns on old, lower leaves?
- Are leaves mottled light and dark green, twisted, or thin and fern-like?
- Has it been cool, cloudy, rainy, foggy, or humid? Does water fall on the leaves when watering?
- Are volunteer tomato or potato plants, infected tubers, or similar weeds nearby?
- What sprays have already been used? Ask for the product name, label, date, and amount only to help the expert; do not advise extra application from that information alone.

Act on urgency

- **Fast spread, cool/wet weather, oily/water-soaked patches, white fuzz, dark stem patches, or fruit/tuber lesions:** Treat suspected late blight as urgent. Ask the farmer to contact a local agriculture officer/KVK promptly. Reduce leaf wetness, avoid moving contaminated plant material around, and do not compost suspected late-blight plants.
- **Round target-like brown spots first on older lower leaves, with a yellow edge and slower spread:** Early blight is possible. Remove badly diseased material, reduce leaf wetness, and seek local guidance if many plants or upper growth are affected.
- **Light/dark green mosaic, distorted young growth, small plants, or blotchy fruit:** A virus such as tomato mosaic virus is possible. There is no spray cure in these materials. Prevent spread by cleaning hands/tools and seek confirmation, especially if herbicide injury or another virus may fit.
- **No visible symptoms:** Do not recommend a routine pesticide spray just because the plant is healthy. Encourage scouting and prevention. A local expert may advise protective measures during a local high-risk warning.
- **Image is unclear, symptoms conflict, or top possible diseases are close:** Say that the result is uncertain. Ask for better photos and suggest expert confirmation; do not give a disease-specific chemical plan.

3. Disease quick comparison

Late blight (Tomato and potato)

What it often looks like: Large, irregular, dark/olive or water-soaked greasy patches; pale edge; white/grey fuzz underneath in humid weather; dark wet stem lesions. Tomato fruit or potato tubers can get firm brown sunken patches.

Typical pattern / conditions: Can spread very fast in cool, cloudy, wet and humid conditions. Wind and splashing water move spores.

Key first step: Treat as urgent if suspected. Check the full plant; contact local expert; remove heavily infected plants safely; do not compost them.

Early blight (Tomato and potato)

What it often looks like: Dark brown, round spots with rings like a target, often a yellow edge. Usually starts on oldest, lower leaves. Potato tubers may have dry, corky brown rot.

Typical pattern / conditions: Often develops more slowly than late blight; rain, dew, wet foliage, warm conditions, and stressed plants can help it.

Key first step: Remove badly affected leaves/plant waste; keep leaves dry; improve airflow; ask for local spray advice if needed.

Tomato mosaic virus (Tomato)

What it often looks like: Light and dark green or yellow patches; twisted young leaves; narrow/fern-like leaves; stunted plant; blotchy fruit.

Typical pattern / conditions: Can move by hands, clothes, tools, handling, and seed; weeds/other host plants may matter.

Key first step: No cure spray. Remove suspect plants as advised; clean hands/tools; do not save seed from sick plants. Confirm because other problems can look similar.

Healthy appearance (Tomato and potato)

What it often looks like: No obvious disease in the photographed part.

Typical pattern / conditions: A single photo may miss early symptoms elsewhere. Wet/cool weather increases the need to check for late blight.

Key first step: No spray needed from the photo alone. Check the whole plant regularly and after wet weather.

Rule of thumb: Target-like rings on old lower leaves point more toward early blight. Large greasy/water-soaked patches, white fuzz underneath, stem lesions, and fast spread point more toward late blight. Mosaic colour patterns and twisted young leaves may point toward a virus. These are clues, not proof.

4. Healthy tomato and potato plants

What “healthy” means in a photo

The photographed leaf looks healthy. This does not prove that the entire plant or field is disease-free. A photo may miss disease on the leaf underside, another leaf, stem, flower, fruit, or tuber. Do not recommend spray just because the app classified a plant as healthy.

Weekly check (about five minutes)

1. Look at the top and underside of leaves, the stems, and fruit. For potato, inspect the plant and check tubers at harvest.
2. Look for oily dark patches and white fuzz (possible late blight), target-like rings on old lower leaves (possible early blight), and green mosaic or twisted new leaves (possible virus in tomato).
3. Check more often after rain, fog, or cool cloudy weather. Suspected late blight can move quickly in wet conditions.
4. Photograph any new sign clearly, including the whole plant and close-ups of both sides of a leaf, stem, and fruit/tuber.

Prevention habits

- Start with healthy transplants; use clean, trusted seed or seed potatoes. Potato late blight can be carried by infected tubers.
- Give plants and rows enough space for air to move. Stake tomato plants if suitable. Open, airy plants dry faster.
- Water the soil rather than the leaves. Avoid overhead sprinklers where possible. Improve drainage and avoid leaving plants wet for long periods.
- Remove weeds and volunteer tomato/potato plants. They can carry disease. Weed identification and host range may need local advice.
- Rotate crops where practical. Do not plant tomato in the same field every year; use local rotation advice. The mosaic plan recommends at least a two-year break and avoiding some related crops, but this needs local crop planning.
- Use mulch carefully. Source material mentions black polythene around 30 micron and a thin organic layer around one inch in a tomato prevention summary. These are not universal prescriptions; local climate, soil, water, and disposal practices matter. Avoid thick wet mulch piled against plants.
- Keep hands and tools clean, especially when handling a plant suspected of mosaic virus. Work in the affected area last where practical.
- Ask a KVK or agriculture officer about varieties suited to the district. The supplied materials name Arka Abhed F1 for late-blight risk in high-rainfall India; the other resistant varieties listed are from US sources and may not be available or suited locally.
- Follow local weather and disease warnings. A healthy-looking crop may still need extra scouting after a wet spell or nearby outbreak.

Healthy-plant photo follow-up (pilot suggestion)

The supplied plans suggest one photo every 7 days, plus an extra photo within 2-3 days after rain or a cool cloudy spell. This is a project-team schedule, not a validated agricultural rule. The chatbot may use it as a reminder suggestion but should label it as such and adapt to local expert advice.

5. Late blight: tomato and potato

What it is

Late blight is caused by *Phytophthora infestans*, a water mold (a fungus-like organism). It affects tomato and potato. It can attack leaves, stems, flowers, and fruit, and potato tubers. It is not the same disease as early blight. It is most dangerous when the weather is cool, cloudy, wet, and humid.

What to look for

Leaves: Tiny irregular brown or water-soaked spots may grow into large green, olive, brown, or dark patches. In wet weather they may look oily or greasy, often with a pale green or grey edge. In humid conditions, white fuzzy growth may appear on the underside, especially in shaded areas. In dry weather, marks may look more papery.

Stems and leaf stalks: Brown to black, irregular, wet-looking or greasy streaks/patches. Flowers may darken or drop.

Tomato fruit: Firm golden, olive, or dark brown sunken patches, sometimes with rings or white fuzz. Fruit may soften or rot.

Potato tubers: Firm, brown, sunken patches. Infected tubers can rot in storage and affect other stored potatoes. Do not store or eat tubers with suspicious brown sunken marks; get local advice and keep them away from healthy seed/storage lots.

Speed: The supplied sources say symptoms may appear within about three days after infection and, in cool rainy overcast weather, plants may collapse within roughly 7-10 days. These are source-reported examples, not a reliable countdown for every field.

Weather and spread

- Cool, wet, humid, cloudy weather favours the disease. An Indian article cites about 15-18°C and humidity above 90%; other extension sources describe a broader cool-to-mild temperature range. Do not turn these mixed source thresholds into a hard alert boundary.
- Spores travel in wind and splashing water. Sources say they can travel miles; one cites up to about 40 miles. A nearby outbreak matters even if the farmer has not handled the sick plants.
- Overhead watering, crowded plants, poor airflow, poor drainage, and long wet periods increase risk.
- Infected potato tubers, volunteer tomato/potato plants, cull piles, infected transplants, nightshade weeds, and other living host plants can help carry the disease between crops. Tomato seed is generally not considered a source in the cited extension material.
- The supplied sources disagree or are uncertain about survival in plant debris, soil, and some hosts in different regions. For India, the chatbot should not assert a universal soil-survival rule. It should say to remove volunteer hosts and infected material and ask a local plant-health expert.

What to do now if late blight is possible

1. Look at the whole plant, both sides of leaves, stems, fruit/tubers, and nearby plants. Take clear photos.
2. Contact the local agriculture officer/KVK promptly if the symptoms are spreading quickly, many plants are affected, or stems/fruit/tubers are involved.

3. Remove heavily infected plants or material so they are no longer alive and releasing spores. The source plans suggest bagging and discarding or deep burial; ask local authorities for a safe disposal option. Do not compost infected plants. Some source alternatives (such as solar covering) are not universal, so do not present them as the default.
4. Remove volunteer tomato/potato plants and relevant weeds nearby. Do not move contaminated plants around the field.
5. Keep leaves as dry as possible: stop overhead watering, improve drainage where practical, and allow airflow.
6. Tell nearby growers or the local agriculture service when an outbreak seems likely; wind can carry spores across fields.
7. A protective fungicide may protect healthy leaves before infection, but it will not restore dead leaves and may work poorly after disease is established. Any pesticide decision belongs to the local expert and the current product label.

Prevention for late blight

- Use healthy transplants and certified disease-free seed potatoes.
- Remove cull/waste potatoes and volunteer tomato/potato plants. Do not use suspect tubers as seed.
- Space and stake plants, improve airflow and drainage, keep leaves dry, and scout during cool wet spells.
- Consider resistant/tolerant varieties if a local expert confirms availability and suitability. The supplied Indian article names Arka Abhed F1 for high-rainfall areas. US varieties named in source PDFs (Mountain Magic, Plum Regal, Defiant, Legend) are not confirmed for Indian availability.
- Protective sprays, where locally approved and advised, are preventive tools. Coverage on healthy foliage and new growth matters; rain and plant growth can reduce protection. Do not rely on a spray to rescue a severely infected field.

Late blight on tomato versus potato

The organism and weather pattern are shared. Both crops can show dark, wet-looking leaf and stem lesions and spread disease between nearby crops. Tomato plans focus on fruit lesions; potato plans add infected seed tubers, waste potatoes, tuber rot, and storage risk. If either crop has suspected late blight, also inspect the other crop and notify the local expert.

Tomato late-blight fruit questions

- The FAQ says healthy-looking fruit from an infected plant may be safe to eat, but it may not store well. It advises against canning fruit that shows late-blight lesions. If a fruit has a lesion or the farmer is unsure, recommend discarding it and asking local food-safety/agriculture guidance. Do not guarantee safety based on a photo.
- Do not compost infected plants. Warm compost may not reliably kill the organism.
- Late blight generally is not considered tomato-seedborne in cited extension sources, but infected potato tubers are an important risk.

6. Early blight: tomato and potato

What it looks like

Early blight is associated with *Alternaria solani*. It often begins on the oldest, lowest leaves. Spots are dark brown, round, and develop rings like a target. A yellow edge may appear. Spots can join; lower leaves yellow, die, and fall. It commonly becomes noticeable as the plant canopy gets thick and humid.

- **Tomato:** Spots can occur on leaves and stems. Fruit may develop dark spots near the stem.
- **Potato:** Old leaves show target-like brown spots; badly affected leaves yellow and fall. Potato tubers may develop brown, dry, corky rot.
- Rain, dew, warm humid conditions, or sprinkler water can help infection. Stressed or poorly nourished plants may be more affected; the supplied sources do not provide a general fertilizer amount.

How to tell it from late blight

Early blight more often starts low on old leaves and has small, round target rings. Late blight more often has larger irregular, greasy/water-soaked patches, possible white fuzz underneath, stem lesions, and rapid spread in cool wet weather. This comparison is not enough to confirm disease. Septoria leaf spot and frost injury can also resemble late blight; the supplied materials have limited information on those look-alikes. Ask an expert if unsure.

What to do

1. Remove selected sick leaves/branches to improve airflow; remove and destroy badly sick plants and crop leftovers where appropriate.
2. Remove weeds and volunteer tomato/potato plants.
3. Keep leaves dry; avoid overhead watering. Stake/space tomato and keep potato rows open where practical.
4. Maintain good crop growth with appropriate water and plant food, but ask a local expert for amounts suited to the soil and crop stage.
5. Watch fruit and potato tubers as well as leaves. Remove suspect tubers from storage/seed lots and ask an expert.
6. If marks are oily, have white fuzz, reach stems/fruit, or spread very fast, do not assume it is early blight. Treat late blight as a possibility and seek urgent local advice.

Chemical information in supplied early-blight plans: not a chatbot prescription

The tomato plan reports a research-trial rate of mancozeb 75% WP at 2 g/L (the document calculates 30 g in 15 L). It also mentions copper and chlorothalonil without a rate. The potato plan reports zineb or captan at 2 g/L and a 15-20 day interval from a cited source; it mentions other products and conflicting schedules. The source documents explicitly say the rates came from research/advisories, not current labels, and Indian product approvals were not checked. A US extension schedule differs. Therefore:

- Do not turn these figures into direct spray instructions in an answer.
- If asked about them, explain they are historical/source-reported research figures, not a verified local recommendation, and send the farmer to the current label and local KVK/agriculture officer.
- Do not recommend use if the current local label does not list the crop and disease, or if local regulations/expert advice do not allow it.

• Never guess repeat interval, total applications, protective equipment, or pre-harvest waiting days.

7. Tomato mosaic virus

What it may look like

Tomato mosaic virus (ToMV) can cause light and dark green or yellow patches on leaves. Young leaves may twist or become narrow and fern-like. Plants may stay small and yield less. Fruit may have yellow or brown blotches.

The look can be confused with other viruses or weed-killer injury. The supplied material says weed-killer damage may appear soon after spraying and affect other broad-leaved plants nearby too. A lab test is needed for certainty. Do not diagnose a virus from leaf colour alone.

Important fact: no cure spray

The supplied documents say no spray cures a plant infected with a virus. Do not encourage spending money on fungicides or insecticides as a cure for ToMV. Some other tomato viruses can be spread by insects, so first confirm which virus/problem is present before suggesting insect control.

What to do if mosaic virus is suspected

1. Ask a local agriculture officer/KVK to confirm if the problem is uncertain, spreading, or also seen in seedlings.
2. Remove clearly sick plants as soon as practical to reduce plant-to-plant spread; put removed material in a closed bag and follow local disposal advice.
3. Wash hands and tools with soap and clean water after touching a suspect plant and before handling healthy plants. Clean tools between plants. Work in the affected area last when practical.
4. Avoid unnecessary touching, clipping, tying, pruning, or handling seedlings because handling can move virus.
5. Do not save seed from sick plants. Use certified clean seed or seed from healthy plants; ask an expert before any seed treatment.
6. Remove relevant weeds and volunteer host plants. The supplied plan recommends crop rotation, including a two-year interval, but crop rotations and host range need local advice.
7. Do not apply a stronger cleaner or chemical unless an expert and its label direct it. One source mentions a seed-soak method using trisodium phosphate; the plan itself says not to do it without expert advice. The chatbot must not provide a home recipe or concentration.

Mosaic-virus follow-up (pilot suggestion)

The source plan suggests photos on Day 0, 3, 7, 10, and 14, then weekly. If a few new plants become sick, remove suspect plants and clean hands/tools again, then contact an expert. If many new plants show symptoms, contact an expert promptly. This calendar was suggested by the project team and has not been validated by an expert.

8. Sprays, pesticide safety, and legal limits

This section is intentionally strict because incorrect pesticide advice can harm people, crops, beneficial organisms, and the marketability of produce.

Chatbot pesticide rules

- First say whether a spray is actually relevant. Viruses do not have a cure spray in these materials. A healthy-looking plant does not need an automatic spray.
- Do not state a product dose as an instruction unless the current local product label confirms it for this crop and disease and a local qualified adviser agrees.
- The supplied PDF doses are not verified against current Indian registration or product labels. Research rates are not automatically legal or suitable rates for a farmer's product.
- Never combine products, change amounts, shorten intervals, or extend a schedule on your own. Do not infer a 15-litre pump amount unless an expert has directed it and the label supports it.
- Never invent the number of sprays or the days to wait before harvest. The label is authoritative for the product's approved use and waiting time.
- If a farmer has already sprayed, do not recommend another application without the product name, label, date, local rules, and expert review. Refer them to an agriculture officer/poison-control or medical service if there was exposure or illness.
- If approved spraying is advised, repeat the label's protective clothing and handling rules in simple language. Keep people, animals, food, and water sources away as the label directs; store pesticide in its original container, locked away from children and food. Do not eat, drink, or smoke while mixing or spraying. Wash after use. Do not give an improvised disposal method.
- Do not recommend banned or unregistered products, even if they appear in an old study or an overseas extension page.
- Fungicides used repeatedly in the same group can lose effectiveness as resistance develops. Only a local expert should select and rotate approved groups; do not invent a rotation or tank mix.

Research figures mentioned in the source PDFs (for traceability only)

These numbers are preserved because they appear in the supplied material, but they are **not verified pesticide directions** and should not be served by the chatbot as a recommendation:

Source-plan context	Figure reported in the PDF	Why the chatbot must not prescribe it
Tomato late blight	Copper oxychloride 50% WP: 2.5 g/L; source trial described six sprays at six-day intervals with black plastic mulch.	Trial/advisory figure, not confirmed against current label, local approval, crop conditions, or harvest waiting period.
Tomato late blight	Copper hydroxide: one source table reports 2.0 g/L; a cited trial reports 0.25% (2.5 g/L).	The PDFs themselves flag the rate conflict. Label and local expert must decide.
Potato late blight	Mancozeb 75% WP: 2 g/L; then cymoxanil 8% + mancozeb 64% WP: 3 g/L at 7-10 day intervals; optional sticker 1 mL/L in the reported plan.	Source is a research trial/news report of an advisory; local approval and labels unverified. Never infer or prescribe the sequence.
Tomato early blight	Mancozeb 75% WP: 2 g/L reported from a field trial.	Trial rate only; current product approval and label not checked.
Potato early blight	Zineb or captan: 2 g/L; source reports every 15-20 days.	Old/source-reported rate; another cited jurisdiction has a different schedule; approval and label unverified.

Source-plan context	Figure reported in the PDF	Why the chatbot must not prescribe it
Potato healthy/high risk	Mancozeb 0.2% (2 g/L) is reported via a news report of a CPRI advisory.	The original advisory was not included; do not convert to advice without obtaining and checking it.

Do not use this table to mix or spray pesticide. It is included so the chatbot team knows exactly which figures exist and why they are gated. A local agricultural expert should either replace these entries with verified current recommendations or keep them unavailable to farmer-facing answers.

9. When to contact an expert

Recommend a local KVK, district/block agriculture officer, state agricultural university, or qualified plant-health professional when:

- The chatbot is unsure, the image is poor, or two possible diseases are close.
- Late blight is possible, especially after cool, wet weather or a nearby outbreak.
- Symptoms spread over a few days, many plants are affected, or the crop is collapsing.
- Marks occur on stems, flowers, fruit, or potato tubers.
- Potato tubers show brown sunken spots or a store may be affected.
- A suspected virus is spreading, seedlings are affected, or herbicide injury is possible.
- The farmer wants to buy/apply a pesticide, asks for a dose or harvest waiting period, is growing organically, or sprays do not seem to work.
- The farmer is unsure whether damaged produce can be eaten, stored, sold, or processed.

Suggest that the farmer prepare:

- Clear photos: whole plant, leaf top, leaf underside, stem, fruit/tuber, and nearby affected plants.
- Crop name/variety if known, field location, crop age, and when symptoms first appeared.
- How many plants are affected and whether symptoms changed over 2-3 days.
- Recent rain, fog, humidity, irrigation method, nearby tomato/potato crops, and any spray details (product and label).

The supplied PDF names KVK and agriculture officers as useful Indian contacts. It says phone numbers/contact lists must be verified before release; do not fabricate contact numbers or locations.

10. Follow-up photo schedules from the supplied plans

These are **pilot reminders**, not expert-validated treatment schedules. Never make a pesticide application depend only on an app calendar. Local label and expert advice take precedence.

Case	Suggested photo checks in source plan
Healthy tomato / healthy potato	Every 7 days; extra photo within 2-3 days after rain or a cool cloudy spell.
Tomato late blight	Day 0, Day 3, then Day 6/9/12/18/24/30 and weekly afterward, especially after rain/cloud. The source also lays out spray dates; do not treat those spray dates as verified instructions.
Potato late blight	Day 0, 3, 6, 8, 11, 14, 16, then weekly and after rainy/cloudy spells. The source's day 8/day 16 spray entries are not label-verified directions.
Tomato early blight	Day 0, 5, 10, 15, 20, 30, then weekly. Spray timing in the source plan is not a verified local schedule.
Potato early blight	Day 0, 5, 10, 15, 20, 30, then weekly. Spray timing in the source plan is not a verified local schedule.
Tomato mosaic virus	Day 0, 3, 7, 10, 14, then weekly. Project-team suggestion; ask expert if new cases appear.

For every new image, compare the same plant/area where possible and ask whether new plants have developed symptoms. An unchanged photo does not prove the disease is gone. If more plants are affected, symptoms move to stems or fruit/tubers, or the crop worsens, recommend expert help promptly.

11. Frequently asked questions: simple answers

Can a photo prove the disease? No. A photo can suggest a disease. Similar problems can look alike. Check the whole plant and ask an expert when unsure.

Can a spray cure tomato mosaic virus? No cure spray is listed in these materials. Remove sources of spread, clean hands and tools, and ask an expert to confirm the problem.

Can a fungicide bring dead leaves back? No. A fungicide may protect healthy leaves when used correctly, but it does not restore dead tissue. Late-blight sprays work best before infection and may work poorly after disease has spread.

Should I spray a healthy-looking plant just in case? Not based on the photo alone. Keep checking the crop. A local expert may advise protection when disease risk is high in your area.

Can late blight spread from another field? Yes. Spores can move in wind and rain splash over long distances. Warn nearby growers and contact the local agriculture service if an outbreak is likely.

Can I compost late-blight plants? No. The supplied sources warn that compost may not heat enough to kill the disease. Bag, bury, or dispose of material only as local guidance permits.

Can I save potato tubers with late-blight marks? Do not store them with healthy potatoes or keep them as seed. They can rot and damage the store. Ask a local expert how to dispose of them.

Can I eat a tomato from an infected plant? The source FAQ says fruit that looks healthy may be safe to eat but may not store well. Do not eat, store, or can fruit with lesions based only on chatbot advice; discard visibly affected fruit and ask local food-safety/agriculture guidance when uncertain.

Does late blight come from tomato seed? The cited extension sources generally say it is not seedborne in tomato. Infected potato tubers and living volunteer plants are more important risks in those sources.

What weather makes late blight worse? Cool, cloudy, humid, rainy weather and leaves that stay wet. Hot, dry, sunny weather usually slows it, but wet leaves from overhead watering can still raise risk.

How can I tell early and late blight apart? Early blight often makes small round spots with rings on old lower leaves. Late blight often makes larger greasy wet-looking patches, white fuzz underneath, stem spots, and fast spread. An expert may need to confirm.

Will it stay in my field next year? Late blight commonly survives on living infected plant material such as volunteer plants or potato tubers, but the supplied sources differ about survival in debris/soil in different places. Remove volunteer hosts and ask about local risk.

Which variety should I plant? Ask the local KVK or agriculture officer which varieties are available and suited to your area. The source set mentions Arka Abhed F1 for high-rainfall Indian conditions; it does not verify local availability or suitability in every district.

What if the app says it is not sure? Take clear photos of both sides of a leaf, stem, and fruit/tuber. Do not spray based on a guess. Contact an expert if symptoms are spreading or severe.

12. Chatbot response examples

Example A: suspected late blight on tomato

“Your tomato may have late blight because the marks look wet and dark. I cannot confirm this from one photo. This disease can spread quickly in cool, rainy weather. Please check under the leaves and look at the stems and fruit. Keep water off the leaves. Do not put sick plants in compost. If many plants are affected or the marks are spreading, contact your local KVK or agriculture officer today. Do not spray until an expert checks the product and its label for your area.”

Example B: target-shaped spots on potato leaves

“The round spots with rings on the older leaves may be early blight. Remove badly sick leaves and crop waste, and avoid watering over the leaves. Check whether tubers also have dry brown marks. A photo cannot confirm the disease. Ask your agriculture officer before choosing a spray; the rates in the guide are not verified product-label instructions.”

Example C: mosaic and curled new tomato leaves

“The light and dark patches and twisted young leaves may be a virus, but weed-killer damage or another problem can look similar. There is no cure spray for tomato mosaic virus in this guide. Wash your hands and tools after touching the plant, avoid handling healthy plants next, and do not save seed from this plant. Please ask your KVK or agriculture officer to confirm.”

Example D: healthy-looking tomato

“The leaf in this photo looks healthy. A single photo cannot check the whole plant. No spray is needed from this result alone. Look at both sides of leaves, stems, and fruit once a week, and check again after rain or cool cloudy weather.”

13. Source review notes and limits

The supplied documents are a pilot knowledge set, mostly version 0.1, dated 11 October 2026, and labelled **DRAFT - not yet expert-checked**. They were prepared by a CropCare AI team with AI-assisted research. The chatbot team should retain that status in its records.

Important unresolved or conflicting points retained from the source PDFs

- The tomato mosaic plan's model-confidence threshold (70%) and photo calendar are project-team suggestions, not validated thresholds.
- Early-blight pesticide rates are trial/advisory rates, not verified against Indian registrations or current product labels. The potato early-blight material also notes that a US repeat interval differs from an Indian source.
- Potato late-blight rates rely partly on a news report of a CPRI advisory; the original advisory was not included. The product labels/Indian approvals are unverified.
- Tomato late-blight copper hydroxide is listed at two different rates (2.0 g/L and 0.25%, or 2.5 g/L) in the source set.
- Late-blight survival between crops differs by region and source. Sources agree that living infected tissue and potato tubers/volunteer plants are important but are not fully aligned on debris/soil survival.
- Host lists vary for pepper and eggplant. This guide focuses on tomato and potato and does not claim those other crops are confirmed hosts in every region.
- Late-blight look-alikes include early blight, Septoria leaf spot, and frost injury, but the supplied look-alike treatment is limited. Add locally reviewed diagnostic material before the chatbot handles more diseases.
- The healthy-plant plans are not complete crop-care manuals. They do not give general fertilizer, irrigation, pest, or storage recommendations.
- Contact information for local Indian services was not verified in the supplied set. Do not invent phone numbers.
- One source suggests diluted bleach cleaning for stakes/cages; another says late blight does not persist on stakes. The chatbot should not give bleach instructions as a universal requirement. Use cleaning advice only when a local expert and product label support it; never mix cleaners.
- Mosaic seed treatment using trisodium phosphate appears in a source, but the plan itself says not to do it without expert advice. The chatbot must not provide a home recipe or concentration.

Validation needed before deployment

1. Have an Indian plant pathologist/agronomist review each diagnosis, host, and disposal recommendation for the target states/districts.
2. Check every pesticide product against current Indian registration and the exact label, including crop, disease, rate, personal protection, repeat interval, maximum applications, re-entry, and pre-harvest waiting period. Replace or remove every unverified figure.
3. Obtain the original CPRI potato late-blight advisory and review the cited trials/advisories in full.
4. Test the chatbot on poor photos, mixed infections, early/late blight look-alikes, herbicide injury, and user requests for unsafe chemical combinations.
5. Add verified local contact details and a clear escalation pathway.
6. Validate the model's uncertainty handling and confidence threshold with real field images before using any numeric cutoff.
7. Translate and field-test farmer-facing answers with local farmers; check that instructions are understandable and not too long.

8. Put a visible “possible diagnosis, not confirmed” note in the app, and make expert escalation easy.

14. Supplied PDF inventory used in this consolidation

The guide consolidates these 16 named files provided in the request. Their original citations and source bibliographies should be retained with the project audit trail.

General / simple action plans

1. `tomato_mosaic_virus_10_simple_action_plan.pdf` - tomato mosaic virus actions, hygiene, seed and follow-up.
2. `tomato_healthy_10_simple_action_plan.pdf` - healthy tomato scouting and prevention.
3. `tomato_early_blight_10_simple_action_plan.pdf` - tomato early blight symptoms, actions, source-reported chemical figures and caveats.
4. `potato_late_blight_10_simple_action_plan.pdf` - potato late blight actions, tuber/storage risk, follow-up and source-reported chemical figures.
5. `potato_healthy_10_simple_action_plan.pdf` - healthy potato scouting and prevention.
6. `potato_early_blight_10_simple_action_plan.pdf` - potato early blight symptoms, actions and source-reported chemical figures.
7. `tomato_late_blight_10_simple_action_plan.pdf` - tomato late blight immediate actions, source-reported protection plan, fruit note and follow-up.

Tomato late-blight knowledge modules

8. `tomato_late_blight_01_overview.pdf` - disease cause, host scope and key facts.
9. `tomato_late_blight_02_symptoms.pdf` - symptoms on leaves, stems, flowers, fruit and progression.
10. `tomato_late_blight_03_lookalikes.pdf` - comparison with early blight, Septoria and frost injury.
11. `tomato_late_blight_04_causes_spread.pdf` - weather, spore movement, disease cycle and between-crop survival limits.
12. `tomato_late_blight_05_immediate_actions.pdf` - removal, disposal, moisture control, severity notes and fruit guidance.
13. `tomato_late_blight_06_prevention.pdf` - healthy plants, field practices, varieties and weather watch.
14. `tomato_late_blight_07_chemical_ipm.pdf` - pesticide classes, trial figures, resistance and legal/safety caveats.
15. `tomato_late_blight_08_when_to_consult.pdf` - expert escalation triggers and information to prepare.
16. `tomato_late_blight_09_faq.pdf` - common late-blight questions and answers.

Provenance statement

This guide reorganizes and simplifies the supplied PDFs; it is not an independent laboratory diagnosis, a complete crop-production manual, or a pesticide label. Where the PDFs conflict or lack local verification, the guide preserves uncertainty and directs the chatbot to expert review. The original PDFs should remain available for detailed bibliographies and source-by-source auditing.

15. Set this file up for chatbot retrieval

This file is designed to be the chatbot's only knowledge file. Keep the headings and disease names when uploading it; they help the chatbot find the right section. If the chatbot platform offers a chunk size, use chunks that preserve a complete heading and the paragraphs or list immediately under it. Do not split a pesticide warning away from the pesticide figures it limits.

Retrieval rules

- First identify crop: tomato or potato. Then retrieve that crop's disease section and the shared triage/safety rules.
- For suspected late blight, retrieve both the late-blight section and pesticide safety section, even if the user asks only “what should I spray?”
- For suspected early blight, retrieve the early-blight section, look-alike rules, and pesticide safety section before answering a chemical question.
- For mosaic patterns, retrieve the tomato mosaic section. Do not mix it up with early or late blight. The guide does not provide a cure spray for a virus.
- For a healthy result, retrieve healthy-plant care and uncertainty rules. Do not return pesticide information unless the farmer asks and an expert has verified local guidance.
- If multiple diseases fit, retrieve the quick comparison and ask for better photos or local expert confirmation.
- Never answer a dose question from memory or from a partially retrieved chunk. Retrieve the complete safety section and advise checking the current local label and expert.

Suggested answer fields for the app

The app can store each result as: **crop; possible problem; visible signs; confidence status (uncertain / possible / needs expert); actions now; what not to do; expert trigger; next photo to take; date; prior photo/result; sprays already reported by farmer.** Do not store an unverified model score as a confirmed diagnosis. Do not claim a farmer followed an action unless the farmer said so.

Search words and farmer expressions

Use these words to help retrieve the right section. They are search hints, not extra diagnoses:

- **Late blight:** late blight, potato blight, tomato blight, wet rot, oily leaf marks, water-soaked marks, white fuzz under leaf, grey fuzz, fast plant collapse, brown sunken tomato, brown sunken potato, rainy-weather blight.
- **Early blight:** early blight, target spots, rings on leaf, bullseye spots, brown round spots on old leaves, yellow edge, dry corky potato tuber.
- **Tomato mosaic virus:** mosaic, mottled leaf, light green and dark green patches, twisted new leaves, fern-like leaf, stunted tomato, blotchy tomato, ToMV, virus.
- **Healthy care:** healthy tomato, healthy potato, weekly check, scouting, clean seed, seed potato, row spacing, leaf wetness, volunteer plants, crop rotation.
- **Uncertain / urgent:** not sure, unclear photo, many plants, spreading fast, stem mark, fruit mark, tuber spot, nearby outbreak, spray did not work.

16. One-minute field action cards

These cards are short retrieval units. They do not replace the detailed sections above.

If suspected late blight

Look for: greasy/wet dark patches; white fuzz underneath in humid weather; dark stem marks; fast spread; tomato fruit or potato tuber lesions. **Do now:** inspect the whole plant; keep leaves dry; do not compost sick plants; remove severely infected material using locally accepted disposal; contact KVK/agriculture officer promptly if spread is fast or many plants are affected. **Do not:** promise a cure; recommend an unverified spray rate; store suspicious potato tubers with healthy seed or eating potatoes.

If suspected early blight

Look for: round brown target rings, usually starting on old lower leaves; often a yellow edge; slower spread than late blight. **Do now:** remove badly affected leaves/waste; keep foliage dry; improve airflow; check fruit or tubers too. **Do not:** assume early blight if lesions are greasy, very fast-spreading, on stems, or have white fuzz underneath; get local help.

If suspected tomato mosaic virus

Look for: light/dark green or yellow mosaic, twisted/narrow young leaves, stunting, blotchy fruit. **Do now:** clean hands and tools; handle affected plants last; do not save seed from sick plants; ask an expert if uncertain or spreading. **Do not:** sell a “cure spray”; give a home seed-treatment recipe; assume every mosaic pattern is ToMV.

If the plant appears healthy

Do now: say the photographed part looks healthy; ask the farmer to check the whole plant weekly and after wet weather. **Do not:** say the field is disease-free; suggest a just-in-case pesticide spray from one healthy photo.

---

17. Plain-language glossary

- **Disease sign:** A mark or change that may show a plant has a problem.
- **Fungus-like water mold:** The organism that causes late blight. It is not exactly a fungus, even though some plant sprays are called fungicides.
- **Spore:** A tiny part of the disease that can spread to another plant, often by wind or splashing water.
- **Volunteer plant:** A tomato or potato plant that grows by itself from a missed seed or tuber.
- **Seed tuber:** A potato saved or bought for planting, not necessarily a botanical seed.
- **Protective spray:** A product that may protect healthy plant surfaces when approved and used as its label says. It does not bring dead leaves back.
- **Pre-harvest waiting time:** The number of days the product label says to wait between spraying and picking. The chatbot must never guess this number.
- **KVK:** Krishi Vigyan Kendra, a local farm science/advisory centre in India.

- **Expert-confirmed diagnosis:** A diagnosis checked by a qualified plant-health professional or lab, rather than guessed from a photo alone.

18. Consolidated references from the supplied PDFs

The following source publications were named in the bibliographies of the supplied PDFs. The references are included here so this PDF can stand alone as the chatbot's knowledge file. Their inclusion does not mean every recommendation was independently rechecked. Some URLs may lead to general pages rather than the exact study record; an expert should review the original references and product-label status before deployment.

1. Cerkauskas R. 'Tomato Mosaic Virus (ToMV)'. AVRDC - The World Vegetable Center, Publication 04-609, 2004. https://worldveg.tind.io/record/39552/files/f0138.pdf
2. University of Minnesota Extension. 'Tomato viruses'. https://extension.umn.edu/disease-management/tomato-viruses
3. Utah State University Extension. 'Tobacco Mosaic Virus (TMV) / Tomato Mosaic Virus (ToMV)'. https://extension.usu.edu/vegetableguide/tomato-pepper-eggplant/TMV-ToMV
4. Meadows I. 'Tobamoviruses that affect tomato: TMV, ToMV, ToBRFV'. NC State Extension. https://content.ces.ncsu.edu/tobamoviruses-that-affect-tomato-tmv-tomv-tobrfv
5. UF/IFAS Extension St. Lucie County. 'Tomato Mosaic Virus (ToMV) and its Management', 3 Mar 2023. https://blogs.ifas.ufl.edu/stlucieco/2023/03/03/tomato-mosaic-virus-tomv-and-its-management/
6. UW Vegetable Pathology (A. Gevens lab), University of Wisconsin-Madison. 'Tomato Late Blight'. Web page, last updated 24 Aug 2026. https://vegpath.plantpath.wisc.edu/diseases/tomato-late-blight/
7. UMass Amherst, Center for Agriculture, Food, and the Environment. 'Late Blight on Tomatoes and Potatoes'. Fact sheet, last updated 4 Apr 2012. https://www.umass.edu/agriculture-food-environment/home-lawn-garden/fact-sheets/late-blight-on-tomatoes-potatoes
8. Singh R. 'Late blight disease of tomato: A serious problem for Indian tomato farmers'. Indian Horticulture (ICAR), Jan-Feb 2026, pp. 34-35. https://epubs.icar.org.in/index.php/IndHort/article/view/164935
9. UC IPM, University of California. 'Tomato Pest Management Guidelines: Late Blight'. https://ipm.ucanr.edu/agriculture/tomato/late-blight/
10. UW Vegetable Pathology, University of Wisconsin-Madison. 'Tomato Early Blight'. Web page, last updated 24 Aug 2026. https://vegpath.plantpath.wisc.edu/diseases/tomato-early-blight/
11. Dushyant, Khatri N.K., Prasad J., Maheshwari S.K. 'Efficacy of fungicides against early blight of tomato caused by Alternaria solani'. 2014 (AGRIS record). https://agris.fao.org/search/en/records/65dfd3c10f3e94b9e5dc9994
12. Pacific Pests, Pathogens and Weeds (PestNet). 'Tomato early blight (Alternaria solani)'. Fact sheet 211. https://apps.lucidcentral.org/ppp_v9/pdf/web_full/tomato_early_blight_211.pdf
13. McGovern R. 'Early Blight of Tomato'. University of Florida IFAS, Disease Management sheet. https://ipm.ifas.ufl.edu/pdfs/Early_Blight.pdf

14. UW Vegetable Pathology, University of Wisconsin-Madison. 'Potato Late Blight'. Web page, last updated 28 Aug 2026. https://vegpath.plantpath.wisc.edu/diseases/potato-late-blight/
15. Lal M., Yadav S., Singh B.P. 'Efficacy of New Fungicides against Late Blight of Potato in Subtropical Plains of India'. Journal of Pure and Applied Microbiology, Mar 2017 (ICAR-CPRI authors). https://doaj.org/article/93e7b2a8e48e473786ecc4172b8e3ea2
16. Indian Potato (news site). 'CPRI Issues Advisory to Safeguard Potato Crops from Late Blight Disease Threat' (report of an ICAR-CPRI advisory; date not shown). https://indianpotato.com/?p=515
17. UW-Extension (Oconto County). 'Late Blight - Deadly Tomato/Potato Disease Found in Area'. Vegetable Grower and Home Garden Alert, 2014. https://oconto.extension.wisc.edu/files/2014/01/late-blight-20141.pdf
18. UC IPM, University of California. 'Potato Pest Management Guidelines: Early Blight'. https://ipm.ucanr.edu/agriculture/potato/early-blight/
19. TNAU Agritech Portal, Tamil Nadu Agricultural University. 'Potato: Early blight (Alternaria solani)'. https://agritech.tnau.ac.in/crop_protection/crop_prot_crop%20diseases_veg_potato_2.html
20. Gevens A.J., Seidl A., Hudelson B. 'Managing Late Blight in Tomatoes'. University of Wisconsin-Extension, 7 Aug 2012. https://waushara.extension.wisc.edu/files/2015/09/Late-blightTomato-Fact-Sheet-Aug-7-20121.pdf
21. University of Vermont Extension. 'Late Blight on Tomato and Potato in 2009'. https://www.uvm.edu/vtvegandberry/factsheets/LateBlight2009.pdf
22. ICAR Research Complex for NEH Region, Sikkim Centre. 'Eco-friendly management of tomato late blight using botanicals, bio-control agents, compost tea and copper fungicides'. Indian Journal of Agricultural Sciences, Mar 2020. https://epubs.icar.org.in/index.php/IJAgS/article/download/98525/39124/341377
23. Utah State University Extension. 'Late Blight' (Vegetable Guide: tomato, pepper, eggplant). https://extension.usu.edu/vegetableguide/tomato-pepper-eggplant/late-blight